

CO4-AT1 — SCENARIO-BASED QUESTIONS

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NO. 016

The Software Company's Speed Claim

THE BRIEF

A software company claims that its new application has an average response time of less than 2 seconds. A hypothesis test gives a p-value of 0.004. At $\alpha = 0.01$, determine whether there is sufficient evidence to support the company's claim.

THE SETUP

Population	every response time the application generates, across all user requests
Sample	a random sample of measured response times, used to run the test
Parameter	The population mean response time, μ , measured in seconds

THE HYPOTHESES

$H_0: \mu \geq 2 \text{ seconds}$

$H_1: \mu < 2 \text{ seconds}$

THE EVIDENCE

- Given p-value = 0.004
- Significance level $\alpha = 0.01$
- $0.004 < 0.01 \rightarrow$ the evidence clears the bar

DECISION: REJECT H_0 — The claim holds up.

READING THE RESULT

At the 1% significance level, a p-value this small — 0.004 — would be a rare accident if the true average response time were really 2 seconds or more. The far more likely explanation is that the application genuinely responds in under 2 seconds.

RECOMMENDATION

The company can confidently publish the sub-2-second benchmark in its marketing and SLAs — the result is unlikely to be a statistical fluke.

FINAL CONCLUSION

Since $p\text{-value} < \alpha$, there is statistically significant evidence that the mean response time is below 2 seconds. Therefore, the company's speed claim is supported.

NO. 017

The Advertisement That Might Not Have Worked

THE BRIEF

A digital marketing team launched a new advertisement and wants to know whether it increases the click-through rate. Before the campaign, the rate was 6%. After the campaign, a hypothesis test produces a p-value of 0.12. What should the data science team conclude at the 5% significance level?

THE SETUP

Population	every visitor who could be shown the advertisement
Sample	the visitors observed after the campaign launched
Parameter	the population click-through rate (proportion), p

THE HYPOTHESES

$$H_0: p = 0.06$$

$$H_1: p > 0.06$$

THE EVIDENCE

- Given p-value = 0.12
- Significance level $\alpha = 0.05$
- $0.12 > 0.05 \rightarrow$ the evidence falls short

DECISION: FAIL TO REJECT H_0 — The case is unproven.

READING THE RESULT

A p-value of 0.12 means there's a 12% chance of seeing a lift this large even if the advertisement changed nothing at all. That's too high a chance to rule out coincidence — the campaign has not cleared the bar of statistical significance.

RECOMMENDATION

Don't credit the campaign with a real lift yet. Let it run longer, collect a bigger sample, or test a bolder creative before drawing conclusions.

FINAL CONCLUSION

Because $p\text{-value} > \alpha$, there is not enough statistical evidence to conclude that the advertisement increased the click-through rate.
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NO. 018

The New Model on Trial

THE BRIEF

A machine-learning team develops a new classification model. The old model has an accuracy of 82%, while the new model achieves 85%. A statistical test gives a p-value of 0.01. Explain what the p-value means and whether the team should conclude that the new model performs significantly better.

THE SETUP

Population	every prediction the model could ever make on the underlying data distribution
Sample	the test set used to measure the new model's 85% accuracy
Parameter	the population accuracy (proportion correctly classified), p

THE HYPOTHESES

$$H_0 : p = 0.82$$

$$H_1 : p > 0.82$$

THE EVIDENCE

- Given p-value = 0.01
- Significance level $\alpha = 0.05$ (standard default)
- $0.01 < 0.05 \rightarrow$ the evidence clears the bar

DECISION: REJECT H_0 — The improvement is statistically significant.

READING THE RESULT

In plain terms, the p-value says: if the new model were truly no better than the old one, a 3-point jump in accuracy this large would happen only 1% of the time by chance. That's a slim enough odds that the improvement is best explained as genuine, not luck.

RECOMMENDATION

Move forward with deploying the new model, while also weighing whether a 3-point accuracy gain is large enough, in practical terms, to justify the switch-over cost.

FINAL CONCLUSION

The p-value indicates strong evidence against the null hypothesis at the 5% level.
The new model's higher accuracy is statistically significant, although practical benefits and deployment cost should also be considered.

NO. 019

The Factory's Defect-Rate Defense

THE BRIEF

A factory claims that its defect rate is 2%. After randomly inspecting products, a 95% confidence interval for the defect rate is found to be (2.4%, 3.6%). At the same time, a hypothesis test gives a p-value of 0.01. Interpret both results and explain whether the company's claim should be rejected.

THE SETUP

Population	every product coming off the factory's line
Sample	the randomly inspected batch used to build the interval and run the test
Parameter	the population proportion of defective products, p

THE HYPOTHESES

$$H_0 : p = 0.02$$

$$H_1 : p \neq 0.02$$

THE EVIDENCE

- 95% confidence interval: (2.4%, 3.6%) — the claimed 2% sits outside it entirely
- Given p-value = 0.01, against $\alpha = 0.05$
- Two independent lines of evidence, one verdict

DECISION: REJECT H_0 — The factory's 2% claim is not supported.

READING THE RESULT

The confidence interval and the p-value are two witnesses telling the same story. Every plausible value in the 95% interval sits above 2%, and the p-value (0.01) says the gap is

too large to blame on chance. Whichever method the factory trusts more, the answer doesn't change.

RECOMMENDATION

The true defect rate is more likely 2.4%–3.6%, not 2%. The factory should investigate its production line before the gap between claim and reality shows up in warranty costs or customer complaints.

FINAL CONCLUSION

Because 2% is outside the 95% confidence interval and $p\text{-value} < 0.05$, the evidence strongly contradicts the company's 2% defect-rate claim.

The Interface Redesign's Bottom Line

THE BRIEF

An online shopping company wants to determine whether a new user-interface design increases the average amount spent per customer. The data scientist calculates a 95% confidence interval for the difference in average spending as (₹120, ₹480) and obtains a p-value of 0.002. Interpret the confidence interval and p-value, and provide a business recommendation.

THE SETUP

Population	every customer who could shop under either interface design
Sample	customers observed under the new interface, compared with those under the old one
Parameter	the difference in mean spending, $\mu_{\text{new}} - \mu_{\text{old}}$

THE HYPOTHESES

$$H_0 : \mu_{\text{new}} - \mu_{\text{old}} = 0$$

$$H_1 : \mu_{\text{new}} - \mu_{\text{old}} \neq 0$$

THE EVIDENCE

- 95% confidence interval for the difference: (₹120, ₹480) — entirely positive, ₹0 is nowhere in it
- Given p-value = 0.002, against $\alpha = 0.05$
- Both a real effect and a sizeable one

DECISION: REJECT H_0 — Adopt the redesign.

READING THE RESULT

Because the interval never dips to ₹0, ₹0 difference is effectively ruled out — and the p-value (0.002) says such a gap would show up only 2 times in 1,000 if the redesign changed nothing. This isn't a marginal result; it's a comfortable one.

RECOMMENDATION

Roll out the new interface platform-wide. Expect average spending to rise somewhere between ₹120 and ₹480 per customer, and track spending post-launch to confirm the gain holds at full scale.

FINAL CONCLUSION

The confidence interval indicates a positive increase in average spending, and the very small p-value confirms statistical significance. The redesign is supported by the evidence, subject to continued post-launch monitoring.